

WAD 2025

Topics and Trends

Arash Kashi July 2025

What is going to happen?

- What should a company of our size focus?
- **Data**
 - <https://github.com/docling-project/docling>

NVIDIA

Learn to build Agentic AI workflows for enterprise Application

- Review tour
 - OpenAI Python Library, LangChain, CrewAI, LangGraph
- AgentIQ NVIDIA tools
- MMLU benchmark basically high school exams for benchmarking models.
- Presentors
 - Harshita Seth
 - Ziv Ilan
 - Lavinia Ghita
 - Sergio Perez

NVIDIA

LLM Compression

- Learn to build Agentic AI workflows for enterprise Application
- LLM Model Compression
 - Quantization (reducing the bits used)
 - Pruning (removing layer, neurons, and attention heads etc.) important
 - PPL (remove and see if it would influence the output)
 - Block (multiple layers Importance (cos distance input and output of a block)
 - Happens usually from layers close to output.
 - Distillation
 - A teacher model would teach the student model how to be. Hard and soft distillation.
- MMLU benchmark (high school questions)
- Triton inference server

NVIDIA

Developer Program

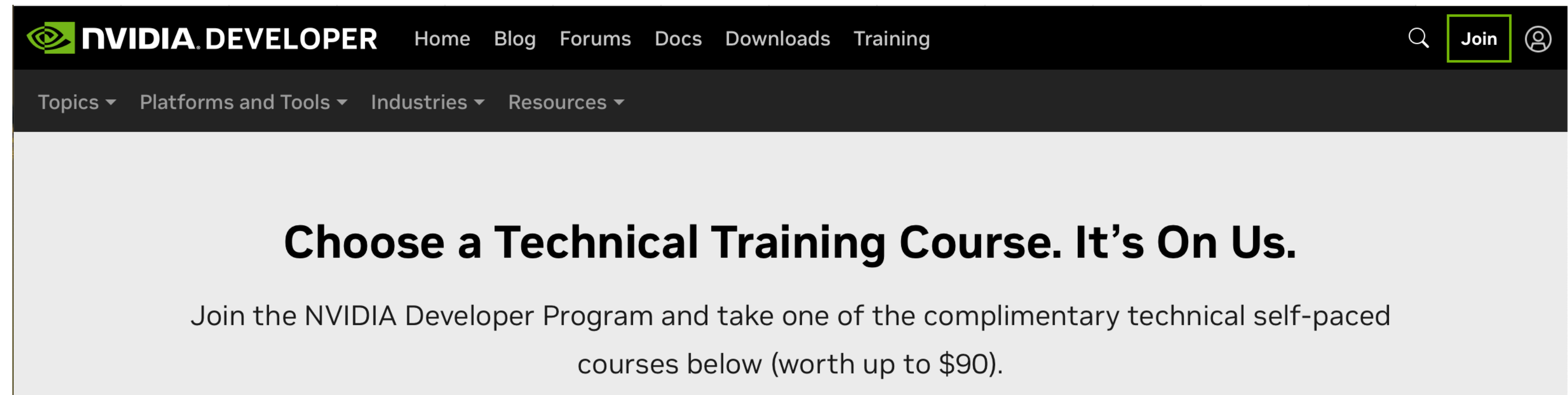
Generative AI and LLMs

Graphics and Simulation

Accelerated computing

Data Science

Deep learning



The screenshot shows the top portion of the NVIDIA Developer Program website. The header is dark with the NVIDIA logo and 'NVIDIA DEVELOPER' text on the left. Navigation links for 'Home', 'Blog', 'Forums', 'Docs', 'Downloads', and 'Training' are in the center. On the right, there is a search icon, a 'Join' button highlighted with a green border, and a user profile icon. Below the header is a dark grey bar with dropdown menus for 'Topics', 'Platforms and Tools', 'Industries', and 'Resources'. The main hero section has a light grey background with the heading 'Choose a Technical Training Course. It's On Us.' and a paragraph: 'Join the NVIDIA Developer Program and take one of the complimentary technical self-paced courses below (worth up to \$90).'

NVIDIA DEVELOPER Home Blog Forums Docs Downloads Training

Topics ▾ Platforms and Tools ▾ Industries ▾ Resources ▾

Choose a Technical Training Course. It's On Us.

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oxylabs.io

Web data scraping

- The "no API" problem
- Turns the entire web into your data source when traditional APIs fall short.

Google

Tools

- aistudio.google.com
 - massive 2M token context windows for processing large documents.
 - Google's answer to OpenAI's playground.
- Agent Development Kit
 - Bidirectional Audio/Video Streaming
 - Agents as Tools for Other Agents (OpenAI, CrewAI)
 - Sub-100 Lines Multi-Agent System
 - [Github \(Jexp/adk-mcp-test\)](#) (Example of ADK)

Siemens

Main Stage Panel (Hanna Hennig CIO)

- How enterprises are including new AI trends.
 - Connect the knowledge of everyone and extrapolate new ideas.
 - Siemens focus on their own data and privacy.
 - How AI could improve products and services? (Interview)
 - Siemens setup their own AI stack securely and provide AI solutions for customers reducing time operating on the same tasks.
 - Find the smallest thing you could apply AI to it, e.g. **excel** sheets.

MCP & RAG

Model Context Protocol

- MCP created out of a hackathon in Anthropic
- Examples
 - RAG = "Give me information to think with" (knowledge retrieval)
 - MCP = "Give me tools to act with" (capability extension)
- AWS MCP Servers
 - <https://awslabs.github.io/mcp/servers/>
- FASTMCP (<https://github.com/jlowin/fastmcp>)

Agents

Tools

- Review of Agent tools
 - LangChain, LangGraph, CrewAI, OpenAI Python Library.
- NeMo Agent Toolkit
- AgentIQ (Agent composer)
- N8N (the future of automation)

Dashboard Tools

- Easy integration.
- AI integration
 - <https://www.dash0.com/>
 - <https://www.netdata.cloud/>

Blockchain

Algorand

- HesabPay in Afghanistan
- Energy efficient, quantum resistant, single layer blockchain with instant finality, consistently high throughput, and low fees, it has been operating at 100% uptime since it went live five years ago.
- Transaction Fee: 0.0002 USD
- Lofty AI
- Gabriel Kuettel
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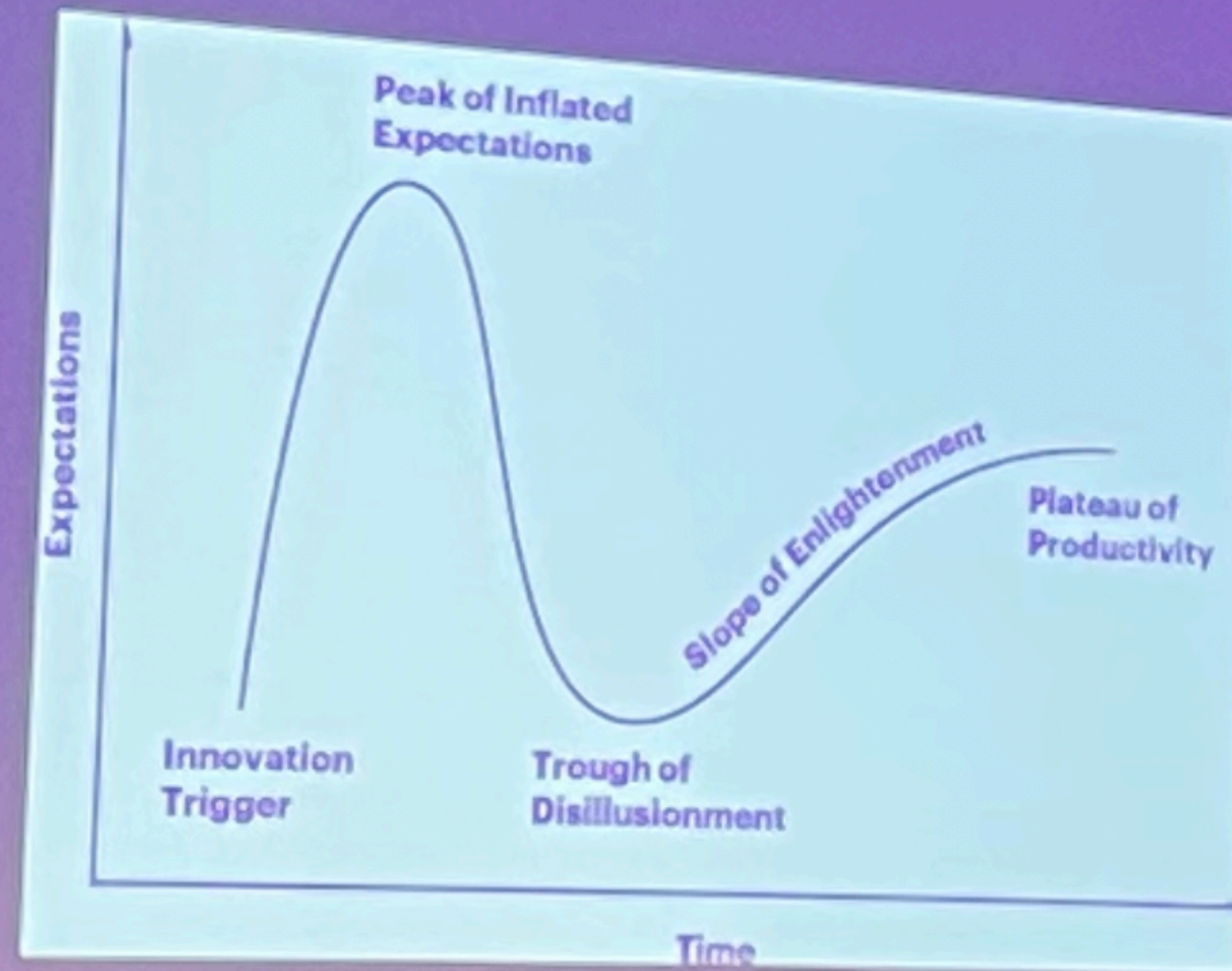
GraphDB

- <https://neo4j.com>





My Hypothesis: All 'Revolutions' follow the same trajectory



> Source: Gartner / BMC

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